Mathematical Errata and Statement Audit

Three papers on chemotaxis systems with logistic sources

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Abstract

This note gives a self-contained mathematical audit of three papers on parabolic–elliptic chemotaxis systems. Each finding is classified as an explicitly false statement, a gap in a printed proof, or a clarification of the intended continuation or regularity framework. False statements are accompanied by explicit counterexamples. For proof gaps, the invalid step, the missing hypothesis, and the range actually supported by the calculation are stated separately. Every conclusion below follows from the displayed mathematical arguments.

Contents

1 Sources and logical classification

The audited sources are:

1. Wenxian Shen, *Existence, uniqueness, stability, and monotonicity of traveling waves for repulsion/attraction chemotaxis models with logistic type source*, arXiv:2605.04401v1 (the "traveling-wave paper");

2. Le Chen, Ian Ruau, and Wenxian Shen, *Chemotaxis models with signal-dependent sensitivity and a logistic-type source, I: Boundedness and global existence*, arXiv:2512.14858v1 ("Part I");

3. Le Chen, Ian Ruau, and Wenxian Shen, *Chemotaxis models with signal-dependent sensitivity and a logistic-type source, II: Persistence and stabilization*, arXiv:2604.02599v1 (“Part II”).

False statement
: The printed hypotheses admit an explicit solution or test object for which the printed conclusion fails.

Proof gap
: A displayed implication is invalid or a hypothesis required by an invoked estimate is missing. This classification does not assert that the theorem itself is false.

Clarification
: The mathematical intent is recoverable, but the literal statement is stronger than the continuation or regularity framework supports.

#	Source	Finding	Status
E1	Traveling wave	Theorems 1.2–1.3: the Section 5 weighted-energy calculation contains a root obstruction, a frame mismatch, and four coefficient defects	PROOF GAP
E2	Traveling wave	Lemma 4.2: the asserted constant subsolution fails for an admissible member of the printed trap set	FALSE STATEMENT
E3	Part I	Theorem 1.2 permits $a > 0, b = 0$, for which spatially constant solutions grow exponentially	FALSE STATEMENT
E4	Part I	Theorem 1.3(iv) invokes an elliptic estimate outside its exponent range	PROOF GAP
E5	Part I	Proposition 1.1 attaches an endpoint alternative to a finite horizon instead of the maximal continuation	CLARIFICATION
E6	Part I	Lemma 2.7 assumes an inequality only for $t > 0$ but concludes a bound including $t = 0$	FALSE STATEMENT as written
E7	Part II	Theorem 2.1(1) permits the pure-decay regime $a = 0 < b$	FALSE STATEMENT
E8	Part II	Theorems 2.2–2.5 claim an all-time C^1 estimate from continuous or merely L^∞-small initial data	CLARIFICATION

2 Traveling-wave paper

2.1 E1: the weighted stability calculation

Let

$$\kappa = \frac{c - \sqrt{c^2 - 4}}{2}, \qquad \kappa^2 - c\kappa + 1 = 0.$$

The printed stability interval begins at $\eta = \kappa$. The coefficient produced by the global estimate in Section 5 is instead

$$q(\eta) = \eta^2 - (c - A)\eta + (1 + B), \tag{1}$$

where the displayed quantities in the paper have $A \geq 0$, and $B > 0$ when $\chi \neq 0$.

Proposition 1 (root obstruction)**.** *Suppose $A \geq 0$, $B > 0$, and the discriminant of q is positive. Then κ is not in the negative set of q. If κ lies to the left of the smaller root*

$$\kappa^- = \frac{(c - A) - \sqrt{(c - A)^2 - 4(1 + B)}}{2},$$

then $q(\eta) > 0$ for every η in a nonempty right-neighborhood of κ.

Proof. Using the defining equation for κ,

$$q(\kappa) = (\kappa^2 - c\kappa + 1) + A\kappa + B = A\kappa + B > 0.$$

Since q is continuous, it remains positive near κ. When $\kappa < \kappa^-$, the quadratic is positive on $(-\infty, \kappa^-)$, negative only between its two roots, and hence the claimed neighborhood is contained in the positive region. $\square$

Consequently, the displayed global coefficient estimate can prove decay only for weights above κ^-, not for every $\eta > \kappa$. This is a gap in the proof: it does not rule out a different localized or spectral argument recovering the larger interval.

Frame mismatch. After the change of variables $z = x - ct$, the estimated energy is

$$E_{\text{mov}}(t) = \int_{\mathbb{R}} e^{2\eta z} \left| u(t, z + ct) - U(z) \right|^2 \, dz.$$

The quantity printed in (1.21) is

$$E_{\text{lab}}(t) = \int_{\mathbb{R}} e^{2\eta x} \left| u(t, x) - U(x - ct) \right|^2 \, dx.$$

The substitution $x = z + ct$ gives the exact relation

$$E_{\text{lab}}(t) = e^{2\eta ct} E_{\text{mov}}(t). \tag{2}$$

Therefore decay of E_{mov} does not by itself imply decay of the printed laboratory-weighted quantity. Either the conclusion must use the co-moving weight $e^{2\eta(x-ct)}$, or the decay rate must be strong enough to absorb the factor in (2). There is also a sign error: if $\lambda = q(\eta) < 0$, then $e^{\lambda t}$, not $e^{-\lambda t}$, is the decaying exponential.

Coefficient audit. Writing $w = u - U$ and $z = v - V$, the elliptic identity $v_{xx} = v - u^\gamma$ gives the zero-order flux difference

$$(va_m - a_{m+\gamma})w + U^m z.$$

After multiplication by $-\chi$, the perturbation equation contains

$$-\chi(va_m - a_{m+\gamma} + b_2)w - \chi b_4 z.$$

Thus the signs of both $a_{m+\gamma}$ and $b_4 z$ in (5.18)–(5.19) must be reversed. Absolute-value estimates can repair those two signs, but three subsequent numerical coefficients must also be changed:

$$|\chi b_1 w_x w| \le \frac{1}{2} w_x^2 + \frac{1}{2} |\chi|^2 b_1^2 w^2, \tag{3}$$

$$\int_{\mathbb{R}} V^2 \le \frac{\gamma^2 M^{2(\gamma-1)}}{(1-\eta)^2} \int_{\mathbb{R}} W^2, \tag{4}$$

$$\int_{\mathbb{R}} V_x^2 \le \frac{\gamma^2 M^{2(\gamma-1)}}{1-\eta^2} \int_{\mathbb{R}} W^2. \tag{5}$$

Equation (3) is Young's inequality with $a = w_x$ and $b = \chi b_1 w$. The factor $M^{2(\gamma-1)}$ in (4)–(5) follows by squaring the Lipschitz bound $|s^\gamma - t^\gamma| \le \gamma M^{\gamma-1} |s-t|$. Finally, an upper bound on U_x/U does not imply an absolute-value bound; the use in (5.23) requires a two-sided estimate for the logarithmic derivative.

Range supported by the corrected calculation. After correcting the flux signs, (3)–(5), and the logarithmic-derivative estimate, the same global quadratic argument supports the window

$$\kappa^- < \eta < \frac{1}{1+|\chi|^{1/6}},$$

provided this interval is nonempty and the corrected discriminant is positive. The energy must be written in the co-moving coordinate. Any claim for the larger interval beginning at κ needs an additional argument not present in Section 5.

2.2 E2: the constant subsolution in Lemma 4.2

The lemma asserts that a sufficiently small constant $W = d$ is a subsolution for every profile in a printed trap set. Consider

$$u(y) = \min\{1, e^{-y/2}\} \min\{1, |y+1|\}.$$

This function is continuous, nonnegative, satisfies the printed upper barrier for $\kappa = \frac{1}{2}$, $M = 1$, and has

$$u(-1) = 0, \qquad u(0) = 1.$$

For $m = \alpha = \gamma = 1$, its frozen elliptic response at -1 is

$$V_0 = \frac{1}{2} \int_{\mathbb{R}} e^{-|-1-y|} u(y)\, dy > 0.$$

Choose

$$\chi = \frac{2}{V_0}, \qquad \kappa = \frac{1}{2}, \quad \tilde{\kappa} = 1, \quad M = 1, \quad c = \frac{5}{2},$$

take D above the printed threshold, and choose a positive d below the printed constant threshold. At $x = -1$, all derivatives of $W = d$ vanish, whereas the frozen operator is

$$\begin{aligned}\mathcal{A}(d;u)(-1) &= -\chi d(V_0 - u(-1)) + d(1-d) \\ &= -2d + d - d^2 = -d(1+d) < 0.\end{aligned}$$

A subsolution requires $\mathcal{A}(d;u) \geq 0$. The universal statement is therefore false.

Proposition 2 (a sufficient repaired condition)**.** *Assume the trap satisfies $0 \leq u \leq M$, so its elliptic response obeys $0 \leq V \leq M^\gamma$. A constant $d > 0$ is a subsolution whenever*

$$|\chi|\, d^{m-1} M^\gamma \leq 1 - d^\alpha. \tag{6}$$

Proof. For a constant test function the diffusion and first-derivative terms vanish. The chemotactic zero-order contribution is bounded below by $-|\chi|\, d^m M^\gamma$, while the logistic contribution is $d(1-d^\alpha)$. Hence

$$\mathcal{A}(d;u) \geq d\big(1 - d^\alpha - |\chi|\, d^{m-1} M^\gamma\big),$$

which is nonnegative under (6). □

For example, when $m = 1$, the simple assumptions $|\chi|\, M^\gamma \leq \frac{1}{2}$ and $0 < d^\alpha \leq \frac{1}{2}$ imply (6).

3 Part I: boundedness and global existence

3.1 E3: Theorem 1.2 needs a reaction guard

The printed assumptions $a, b \geq 0$ include $a > 0, b = 0$. On any smooth bounded Neumann domain, take $c_0 > 0$ and set

$$u(t,x) = c_0 e^{at}, \qquad v(t,x) = \frac{\nu}{\mu} u(t,x)^\gamma.$$

All spatial derivatives and boundary fluxes vanish. The elliptic equation holds, and the parabolic equation reduces to $u_t = au$. Thus this is a positive global classical solution for every m, β, χ_0, but

$$\|u(t)\|_\infty = c_0 e^{at} \longrightarrow \infty.$$

The counterexample even satisfies the small-sensitivity condition by taking $\chi_0 = 0$. Under the standing nonnegativity assumptions, the natural repair is

$$a = 0 \quad \text{or} \quad b > 0,$$

which excludes exactly the branch $a > 0, b = 0$. The more conservative case split used by the printed proof is $(a = b = 0) \vee (a > 0 \wedge b > 0)$.

3.2 E4: Theorem 1.3(iv) uses an inadmissible exponent

Alternative (iv) assumes

$$\beta \geq \frac{1}{2}, \qquad \alpha = 2m + \gamma - 2.$$

Section 5.4 applies Proposition 2.2 with

$$s(P) = \frac{P + \alpha}{\gamma},$$

although that proposition requires $s(P) > 1$. Under the critical identity,

$$s(P) > 1 \iff P > 2 - 2m.$$

The iteration begins at

$$q_* = \max\{1, N\alpha/2\};$$

hence the printed route requires the missing condition

$$\boxed{\max\{1, N\alpha/2\} > 2 - 2m.} \tag{7}$$

The printed hypotheses do not imply (7). Indeed,

$$N = 1, \quad m = \frac{1}{4}, \quad \gamma = \frac{7}{2}, \quad \alpha = 2, \quad \beta = 1$$

satisfies $\alpha = 2m + \gamma - 2$, but

$$q_* = 1, \qquad s(q_*) = \frac{6}{7} < 1.$$

Moreover $(N\alpha - 2)_+ = 0$, so the first smallness disjunct in (1.25) is automatic. Thus Proposition 2.2 is invoked outside its stated domain. Adding (7) repairs this proof route; without it, a different argument is required.

3.3 E5: Proposition 1.1 must use the maximal horizon

A finite endpoint alternative cannot hold for every finite local horizon. When $a, b > 0$, the positive equilibrium

$$u(t, x) = \left(\frac{a}{b}\right)^{1/\alpha}$$

is a classical solution on every finite interval and neither blows up nor approaches zero at its endpoint.

The standard corrected statement is as follows. Let $\mathcal{T}$ be the set of times to which the initial solution can be continued, and put

$$T_{\max} = \sup \mathcal{T}.$$

Uniqueness makes the local solutions compatible on overlaps, so they glue to a solution on $[0, T_{\max})$. If $T_{\max} < \infty$, the continuation criterion implies that at least one controlling norm becomes unbounded or the positive lower bound degenerates as $t \uparrow T_{\max}$. If neither happened, local existence at a terminal slice would extend the solution beyond $T_{\max}$, contradicting its definition. The endpoint alternative therefore belongs to the distinguished maximal continuation, not to an arbitrary finite restriction of it.

3.4 E6: Lemma 2.7 omits the initial trace

The lemma assumes a differential inequality only on $(0, T_{\max})$ but concludes a supremum bound on $[0, T_{\max})$. Let $T_{\max} = 1$ and take

$$u(t, x) = t^{-1}, \qquad 0 < t < 1.$$

For $p = 2$, the mass quantity is $Y(t) = t^{-2}$, and

$$Y'(t) = -2t^{-3} \le -t^{-3}.$$

Thus the interior inequality is compatible with the lemma's nonnegative constants, while

$$\sup_{0<t<1} Y(t) = \infty.$$

No inequality imposed only at positive times can control an omitted initial trace.

The repair is to assume Y extends continuously to $t = 0$, or at least that $Y(0) < \infty$ is part of the data. Scalar comparison may then be applied on $[0, T]$ for every $T < T_{\max}$, with the comparison constant depending on $Y(0)$ and the coefficients in the differential inequality. The classical PDE initial trace supplies exactly this datum.

4 Part II: persistence and stabilization

4.1 E7: Theorem 2.1(1) omits the pure-decay regime

The printed assumptions allow $a = 0 < b$. For $c_0 > 0$, define

$$u(t,x) = \left(c_0^{-\alpha} + \alpha bt\right)^{-1/\alpha}, \qquad v(t,x) = \frac{\nu}{\mu} u(t,x)^{\gamma}.$$

All spatial terms vanish and

$$u_t = -bu^{1+\alpha}.$$

The solution is global, bounded, and strictly positive for every finite time, yet

$$\lim_{t\to\infty} \inf_{x\in\Omega} u(t,x) = 0.$$

The positive persistence conclusion is therefore false in this regime. The case split actually used in the proof is

$$(a = b = 0) \quad \text{or} \quad (a > 0 \wedge b > 0).$$

When $a > 0, b = 0$, a globally bounded positive solution cannot exist because its mass satisfies

$$\frac{d}{dt}\int_\Omega u = a\int_\Omega u,$$

so that branch of an implication restricted to bounded global solutions is vacuous.

4.2 E8: an all-time C^1 estimate cannot follow from C^0 data

Equation (2.12) asserts a C^1 exponential estimate for every $t \geq 0$, whereas Theorem 2.2 assumes only continuous, positive initial data close in L^∞. On $\Omega = (0, 1)$, consider the Neumann data

$$u_{0,N}(x) = u_* + N^{-1/2}\cos(N\pi x).$$

For large N they are positive, have the same prescribed mass as u_*, and satisfy

$$\|u_{0,N} - u_*\|_{L^\infty} = N^{-1/2} \to 0, \qquad \|\partial_x u_{0,N}\|_{L^\infty} = \pi N^{1/2} \to \infty.$$

Thus no L^∞-ball has a uniform C^1 bound at $t = 0$. For a merely continuous datum, the C^1 norm may not even be defined. This is consistent with parabolic smoothing, whose first-derivative estimate has the short-time form

$$\left\|\partial_x e^{t\Delta} f\right\|_\infty \leq Ct^{-1/2}\|f\|_\infty.$$

There are two correct formulations:

1. if the initial perturbation is small in a fractional domain embedded into C^1, sectorial stability gives an all-time strong-norm estimate;
2. for continuous or L^∞-small data, first evolve to any suitable $t_0 > 0$, then obtain exponential C^1 convergence for $t \geq t_0$.

The linear stability threshold is unaffected; only the initial-time regularity assertion changes.

5 Conclusion

Four findings, E2, E3, E6, and E7, are false as literally stated and are settled by explicit counterexamples. E1 and E4 identify precise gaps in printed estimates without asserting that the associated final theorems are false. E5 and E8 are repaired by attaching the conclusions to the correct maximal-continuation and positive-time regularity frameworks. Keeping these logical categories separate is essential: a defective proof step, an overstated norm at $t = 0$, and a counterexample to the conclusion are different mathematical claims.